A type carrying the W-type rules with a propositional computation rule is precisely a homotopy-initial W-algebra

The Univalent Foundations Program

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Editorial note. The counted theorem carries no label in the pinned source; its identity is fixed by the pinned file together with the original number 5.5.5, obtained from the label-number dump by counting the theorem-class environments of section 5.5 of the same file, where the neighbouring labelled result lem:homotopy-induction-times-3 is 5.5.4 and the section begins at its heading at line 566.

Editorial note. The statement is the author's own thm environment at source lines 700-702 and its proof environment starts on line 705, two lines after the statement environment ends, so the pairing is by adjacency and the retained proof range 705-740 is the whole environment.

Editorial note. The retained counters make the counted theorem print as Theorem 5.5.5, matching the original book numbering; the single context span is one line of the author's prose and consumes no counter value.

Editorial note. The retained proof is the author's own, including the commuting diagram of the projection composite; nothing is added to complete it.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-source copy at sources/hott-induction/retained-preamble.tex, referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: a type carrying the W-type rules with a propositional computation rule is exactly a homotopy-initial W-algebra (unlabelled theorem, original Theorem 5.5.5).

Allowed standard prerequisites (editorial).

- W-type rules. Formation, introduction, elimination and a propositional computation rule for W-types, the computation rule asking only for a path rather than a judgemental equality.. They are the two rule systems that the theorem compares at source lines 700-701.
- Homotopy-initial algebra. An algebra is homotopy-initial when for every other algebra the type of homomorphisms out of it is contractible.. It is the hypothesis of the converse direction and the source of the homomorphism used at source lines 714-715.

- Transport in a family of types. A path in the index type of a family induces a map between the fibres by transport, which is natural with respect to the operations of the family.. It is what defines the recursor at source lines 729-737 and what the final calculation consumes.
- Sigma types and projections. The elements of a Sigma type are pairs, and the first projection forgets the second component.. It is the construction of the type C at source line 713 and of the composite at source lines 716-727.

Source reference. Original theorem 5.4.7: sources/hott-induction/book-labelnumbers-first-edition.tex, source lines 1919–1919. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Indeed, we have the following theorem, which is an improvement over original theorem 5.4.7:

Theorem 5.5.5. *The types satisfying the formation, introduction, elimination, and propositional computation rules for* W*-types are precisely the homotopy-initial* W*-algebras.*

Sketch of proof. Inspecting the proof of original theorem 5.4.7, we see that only the *propositional* computation rule was required to establish the h-initiality of $\mathsf{W}_{(x:A)}B(x)$. For the converse implication, let us assume that the polynomial functor associated to $A : \mathcal{U}$ and $B : A \to \mathcal{U}$, has an h-initial algebra (W, s_W); we show that W satisfies the propositional rules of W-types. The W-introduction rule is simple; namely, for $a : A$ and $t : B(a) \to W$, we define $\mathsf{sup}(a, t) : W$ to be the result of applying the structure map $s_W : PW \to W$ to $(a, t) : PW$. For the W-elimination rule, let us assume its premisses and in particular that $C' : W \to \mathcal{U}$. Using the other premisses, one shows that the type $C :\equiv \sum_{(w:W)} C'(w)$ can be equipped with a structure map $s_C : PC \to C$. By the h-initiality of W, we obtain an algebra homomorphism $(f, s_f) : (W, s_W) \to (C, s_C)$. Furthermore, the first projection $\mathsf{pr}_1 : C \to W$ can be equipped with the structure of a homomorphism, so that we obtain a diagram of the form

$$\begin{array}{ccccc} PW & \xrightarrow{Pf} & PC & \xrightarrow{P\mathsf{pr}_1} & PW \\ \downarrow{\scriptstyle s_W} & & \downarrow{\scriptstyle s_C} & & \downarrow{\scriptstyle s_W} \\ W & \xrightarrow[f]{} & C & \xrightarrow[\mathsf{pr}_1]{} & W. \end{array}$$

But the identity function $1_W : W \to W$ has a canonical structure of an algebra homomorphism and so, by the contractibility of the type of homomorphisms from (W, s_W) to itself, there must be an identity proof between the composite of (f, s_f) with $(\mathsf{pr}_1, s_{\mathsf{pr}_1})$ and $(1_W, s_{1_W})$. This implies, in particular, that there is an identity proof $p : \mathsf{pr}_1 \circ f = 1_W$.

Since $(\mathsf{pr}_2 \circ f)w : C((\mathsf{pr}_1 \circ f)w)$, we can define

$$\mathsf{rec}(w, c) :\equiv p_*((\mathsf{pr}_2 \circ f)w) : C(w)$$

where the transport p_* is with respect to the family

$$\lambda u.\, C \circ u : (W \to W) \to W \to \mathcal{U}.$$

The verification of the propositional W-computation rule is a calculation, involving the naturality properties of operations of the form p_*. □